

Dr Oliver Hamelijnck

Research Scientist | ML Engineer | Probabilistic ML & Generative AI
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OVERVIEW

Machine learning and research engineer specialising in **scalable probabilistic ML** and **production AI systems**, with experience spanning **speech recognition**, **generative AI**, **optimisation**, and **large-scale spatio-temporal** modelling. **PhD in Machine Learning** from the **University of Warwick** with publications at **NeurIPS**, **ICML**, **ICLR**, and **AISTATS**. Experienced **building production ML systems** across speech recognition, forecasting, optimisation, and **agentic/RAG applications**.

EXPERIENCE

2025

AI SCIENTIST UNILINK SOFTWARE

October 2025 - Present | London, UK

- **Architected and deployed** multiple production AI systems across anomaly detection, optimisation/scheduling, and RAG workflows used **internally and by enterprise customers**
- Led end-to-end development of agentic AI applications integrating LLM orchestration, retrieval pipelines, structured business data
- Built ML driven **explainable anomaly detection** methods with containerised inference services deployed via **Kubernetes**, supporting scalable enterprise AI workloads.
- Collaborated directly with product, engineering, and business stakeholders to translate operational requirements into production ML systems.
- **Technical lead** for AI initiatives across the organisation and **co-supervise MSc students** from the University of Warwick

2025

MACHINE LEARNING ENGINEER SPEECHMATICS

February 2025 - October 2025 | Cambridge, UK

- Developed and deployed **200M-parameter** multilingual speech-to-text models in **PyTorch**, contributing to a production release delivered under contractual deadlines.
- Led evaluation and benchmarking of multilingual STT models, influencing architectural and deployment decisions across research and engineering teams
- Implemented data preparation pipelines for ingesting **terabytes of audio data**
- Researched and developed ASR models such as **CTC/TDT** to improve latency and accuracy

2023
2025

RESEARCH FELLOW THE UNIVERSITY OF WARWICK

September 2023 - January 2025 | Warwick, UK

- Developed scalable probabilistic inference methods for **spatio-temporal Gaussian process models**, resulting in publications at **NeurIPS** and **ICML**.
- Designed and implemented scalable probabilistic ML systems in JAX for spatio-temporal inference on large geospatial datasets, bridging methodological research with practical deployment considerations.

2021
2021

RESEARCH SCIENTIST INTERN AMAZON

March - August 2021 | Virtual

- Developed **stochastic optimisation models** incorporating forecast uncertainty to improve delivery logistics
- Collaborated with business stakeholders to deploy models in production, resulting in significant cost savings through more robust forecasts

2017
2019

RESEARCH ASSISTANT ALAN TURING INSTITUTE

September 2017-2019 | London, UK

- Researched SOTA machine learning models to forecast air pollution across London resulting in publications in top tier conferences
- **Developed now-casting system** which pulled and processed spatio-temporal data and retrained these models daily. Enabled the Greater London Authority to make evaluate policies in near real-time

2019
2023

2013
2017

EDUCATION

PhD COMPUTER SCIENCE UNIVERSITY OF WARWICK

Scholarship from The Alan Turing Institute

Thesis: Scalable Bayesian Inference for Spatio-Temporal Gaussian Processes

MEng COMPUTER SCIENCE UNIVERSITY OF WARWICK

Graduated with First Class Honours degree

TECHNICAL SKILLS

MACHINE LEARNING

- Probabilistic ML, Gaussian Processes, Variational Inference, Deep Learning, ASR, RAG Systems, Agentic Workflows
- Experience with analysing and processing **terabytes of audio data**, and large volumes of **spatio-temporal geospatial data** (weather, satellite, ground sensors, GIS maps)

SUPERVISION AND PRESENTATIONS

- **Mentored and co-supervised** multiple masters and undergraduate students
- Wealth of experience **delivering presentations** from posters and talks at conferences, to technical presentations both in person and virtually, including Tübingen University (Germany), Amazon Research (virtual), London City Hall (UK).

LANGUAGES & FRAMEWORKS

- Python, **JAX**, **PyTorch**, NumPy, Scikit-Learn

INFRASTRUCTURE

- Docker, Kubernetes, SLURM, Singularity, Milvus, PostgreSQL, PostGIS, Redis, Azure ML

INTERESTS

Bouldering
Breadmaking
Guitar

PUBLICATIONS

Deep Gaussian Processes on Directed Acyclic Graphs

Submitted to NeurIPs 2026: F. Perlino; O. Hamelijnck; A. Johansen; T. Damoulas;

MAGIC: Multi-Agent Generative Intention Coordination

ICLR 2026 Workshop MALGAI: D. Huk; O. Hamelijnck; D. Demiris; T. Damoulas;

Federated Generalised Variational Inference: A Robust Probabilistic Federated Learning Framework

ICML 2025: T. Mildner; O. Hamelijnck; P. Giampouras; T. Damoulas;

Physics-Informed Variational State-Space Gaussian Processes

NeurIPs 2024: O. Hamelijnck; A. Solin; T. Damoulas;

Spatio-Temporal Variational Gaussian Processes

NeurIPs 2021: O. Hamelijnck*; W.J. Wilkinson*; N.A. Loppi; A. Solin; T. Damoulas;. * = Joint first authors

Transforming Gaussian Processes with Normalising Flows

AISTATS 2021: J. Maronãs*; O. Hamelijnck*; T. Damoulas; M. Steel. * = Joint first authors

Non-stationary Non-separable Gaussian Processes

ICML 2020: K. Wang; O. Hamelijnck; T. Damoulas; M. Steel

Multi-resolution Multi-task Gaussian Processes

NeurIPs 2019: O. Hamelijnck; K. Wang; T. Damoulas; M. Girolami